Chen-Li 4.7

2026-08-10

Danus arXiv DOI

J. Chen and J. Li, *Mean Curvature Flow of Surface in 4-Manifolds*, Adv. Math. 163 (2001), 287-309, Theorem 4.7

Chen-Li 4.7

1. Wang (2001) Kähler-Einstein Li-Yang (2014) Zhang (2015) Qiu-Sun (2019/2020) Kähler
2. Chen-Li (2002) Remark 6.7 “ ”
3. Han-Li Han-Li-Sun Li-Sun
4. Chen-Li (2004) Neves (2007) Maslov

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Wang “ ” Chen-Li (2002)

1.

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- Chen-Li 4.7
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- **A** arXiv
- **B** DOI

$$\sup_{\Sigma_t} |A|^2 \leq \frac{C}{T-t}, \qquad t < T.$$

2.

			4.7 /
2000	Chen-Tian, *Moving Symplectic Curves in Kähler-Einstein Surfaces* [R1]	Theorem 2.4 Kähler Theorem 2.5	A
2001	Wang, *Mean Curvature Flow of Surfaces in Einstein Four-Manifolds* [R3]	Theorem A Kähler-Einstein Kähler Huisken	Chen-Li A
2002	Chen-Li, *Singularities of Codimension Two Mean Curvature Flow of Symplectic Surfaces* [R4]	Theorems 1.2, 4.3, 5.1, 6.6 Remark 6.7	“ ” A
2014	Li-Yang, *Symplectic Mean Curvature Flows in Kähler Surfaces with Positive Holomorphic Sectional Curvatures* [R12]	2 Kähler Theorem 1.3 Chen-Li/Wang	Kähler-Einstein A
2014	Li-Yang, *Generalized Symplectic Mean Curvature Flows in Almost Einstein Surfaces* [R13]		MCF B
2015	Zhang, *Remarks on Symplectic Mean Curvature Flows...* [R14]	Li-Yang Kähler Theorem 3.2 /	arXiv [R12] A
2019	Qiu-Sun, *Mean Curvature Flow of Surfaces in a Hyperkähler 4-Manifold* [R15]	Theorem 1.2 Theorem 1.3	White A
2026	Chen-Han-Li-Sun, *Tangent Flows of Symplectic Mean Curvature Flows* [R18]	Type I*	“ ” “ ” B

3.

			/
2009	Han-Li, *Singularities of Symplectic and Lagrangian Mean Curvature Flows* [R5]	Theorem 2.4 $\mathbb{C}^2$ $-2 \leq K \leq 0$ Corollary 2.6	A
2009	Han-Li, *Translating Solitons to Symplectic and Lagrangian Mean Curvature Flows* [R6]	/	A
2011	Han-Li-Sun, *The Second Type Singularities...* [R8]	Theorems 1.1-1.4 Kähler	A
2024	Li-Sun, *Eternal Solutions to Almost Calibrated Lagrangian and Symplectic Mean Curvature Flows* [R16]	Kähler /	B

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2010	Han-Li, *The Mean Curvature Flow Approach to the Symplectic Isotopy Problem* [R7]	Kähler-Einstein	A
2011	Han-Li, *Long Time Existence of the Symplectic Mean Curvature Flow* [R9]	Kähler	arXiv A
2013	Han-Li-Yang, *Symplectic Mean Curvature Flow in* $\mathbb{CP}^2$ [R10]	$\mathbb{CP}^2$	A/B
2014	Han-Li, *The Mean Curvature Flow Along the Kähler-Ricci Flow* [R11]	Kähler-Ricci MCF /	A
2024	Han-Li-Sun, *Gradient Flow for* β*-Symplectic Critical Surfaces* [R17]	β- / β-	gradient-flow MCF A

5.

			/
2004	Chen-Li, *Singularity of Mean Curvature Flow of Lagrangian Submanifolds* [L1]	Calabi-Yau Theorem 1.1 2 Corollary 6.7	A
2007	Neves, *Singularities of Lagrangian Mean Curvature Flow: Zero-Maslov Class Case* [L2]	Theorems A-B Maslov	Maslov “ ” A
2017	Sun-Yang, *Generalized Lagrangian Mean Curvature Flows in Almost Calabi-Yau Manifolds* [L3]	Neves LMCF Corollary 1.1	A
2021	Lambert-Lotay-Schulze, *Ancient Solutions in Lagrangian Mean Curvature Flow* [L4]	blow-down Lawlor neck blow-down	A
2024	Lotay-Schulze-Székelyhidi, *Ancient Solutions and Translators of Lagrangian Mean Curvature Flow* [L5]	blow-down Maslov Brakke	/ A
2025	Li-Luo-Sun, *Type II Singularities of Lagrangian Mean Curvature Flow with Zero Maslov Class* [L6]	Maslov	A

6.

Kähler	$\cos\alpha > 0$		[R1], [R2], [R3]
Huisken	$1/\cos\alpha$	/	[R2], [R4], [R8]
	$\lvert A\rvert^2 \le C/(T-t)$		[R4], [L1]
			[R15]
	$(T-t)\max\lvert A\rvert^2 \to \infty$	/	[R5], [R6], [R8], [R16]
	$\lvert A\rvert^2$, $\lvert H\rvert^2$, $\cos\alpha$		[R7], [R9], [R10], [R12], [R14]

7. 4.7

7.1

1. **Wang Theorem A** Chen-Li Theorem 4.7 Kähler-Einstein
2. **Chen-Li 2002, Remark 6.7**
3. **Qiu-Sun Theorem 1.3** Kähler

7.2

- “ ” 1
- “ ” “ ” White
- Chen-Li 2002 “ ” Wang
-

7.3

Chen-Li/Wang “ ”
White /
Chen-Li 2002 “ ”

8.

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- **Li-Sun 2024 [R16]**
- **Han-Li-Sun 2024 [R17]** β-
- **Li-Luo-Sun 2025 [L6]** Maslov
- **Chen-Han-Li-Sun 2026 [R18]** MCF Type I*

arXiv:2309.16478 2025-10-13

9.

[R1] J. Chen and G. Tian, “Moving Symplectic Curves in Kähler-Einstein Surfaces,” *Acta Math. Sin. (Engl. Ser.)* 16 (2000), 541-548. DOI: 10.1007/s101140000075. ______.

[R2] J. Chen and J. Li, “Mean Curvature Flow of Surface in 4-Manifolds,” *Adv. Math.* 163 (2001), 287-309. DOI: 10.1006/aima.2001.2008.

[R3] M.-T. Wang, “Mean Curvature Flow of Surfaces in Einstein Four-Manifolds,” *J. Differential Geom.* 57 (2001), 301-338. DOI: 10.4310/jdg/1090348113. arXiv:math/0110019.

[R4] J. Chen and J. Li, “Singularities of Codimension Two Mean Curvature Flow of Symplectic Surfaces,” arXiv:math/0208227 (2002). arXiv.

[R5] X. Han and J. Li, “Singularities of Symplectic and Lagrangian Mean Curvature Flows,” *Front. Math. China* 4 (2009), 283-296. DOI: 10.1007/s11464-009-0018-4. arXiv:math/0611857.

[R6] X. Han and J. Li, “Translating Solitons to Symplectic and Lagrangian Mean Curvature Flows,” *Int. J. Math.* 20 (2009), 443-458. DOI: 10.1142/S0129167X09005352. arXiv:0711.4435.

[R7] X. Han and J. Li, “The Mean Curvature Flow Approach to the Symplectic Isotopy Problem,” *J. Eur. Math. Soc.* 12 (2010), 505-527. DOI: 10.4171/JEMS/207.

[R8] X. Han, J. Li and J. Sun, “The Second Type Singularities of Symplectic and Lagrangian Mean Curvature Flows,” *Chinese Ann. Math. Ser. B* 32 (2011), 223-240. DOI: 10.1007/s11401-011-0635-6.

[R9] X. Han and J. Li, “Long Time Existence of the Symplectic Mean Curvature Flow,” arXiv:1107.0829 (2011). arXiv.

[R10] X. Han, J. Li and L. Yang, “Symplectic Mean Curvature Flow in $\mathbb{CP}^2$,” *Calc. Var. Partial Differential Equations* 48 (2013), 111-129. DOI: 10.1007/s00526-012-0544-x.

[R11] X. Han and J. Li, “The Mean Curvature Flow Along the Kähler-Ricci Flow,” arXiv:1105.1200 (2011). arXiv.

[R12] J. Li and L. Yang, “Symplectic Mean Curvature Flows in Kähler Surfaces with Positive Holomorphic Sectional Curvatures,” *Geom. Dedicata* 170 (2014), 63-69. DOI: 10.1007/s10711-013-9867-9. arXiv:1207.5253.

[R13] J. Li and L. Yang, “Generalized Symplectic Mean Curvature Flows in Almost Einstein Surfaces,” *Chinese Ann. Math. Ser. B* 35 (2014), 33-50. DOI: 10.1007/s11401-013-0817-5.

[R14] S. Zhang, “Remarks on Symplectic Mean Curvature Flows in Kähler Surfaces with Positive Holomorphic Sectional Curvatures,” arXiv:1507.06806 (2015). arXiv.

[R15] L.-J. Qiu and J. Sun, “Mean Curvature Flow of Surfaces in a Hyperkähler 4-Manifold,” arXiv:1902.00645 (2019, revised 2020). arXiv.

[R16] X. Li and J. Sun, “Eternal Solutions to Almost Calibrated Lagrangian and Symplectic Mean Curvature Flows,” *J. Math. Anal. Appl.* 531 (2024), Article 127804. DOI: 10.1016/j.jmaa.2023.127804.

[R17] X. Han, J. Li and J. Sun, “Gradient Flow for β-Symplectic Critical Surfaces,” *Ann. Inst. H. Poincaré C Anal. Non Linéaire* 41 (2024), 1083-1116. DOI: 10.4171/AIHPC/100. ________.

[R18] J. Chen, X. Han, J. Li and J. Sun, “Tangent Flows of Symplectic Mean Curvature Flows,” *J. Math. Study* 59 (2026), 40-59. DOI: 10.4208/jms.v59n1.26.03. _______.

[L1] J. Chen and J. Li, “Singularity of Mean Curvature Flow of Lagrangian Submanifolds,” *Invent. Math.* 156 (2004), 25-51. DOI: 10.1007/s00222-003-0332-5. arXiv:math/0301281.

[L2] A. Neves, “Singularities of Lagrangian Mean Curvature Flow: Zero-Maslov Class Case,” *Invent. Math.* 168 (2007), 449-484. DOI: 10.1007/s00222-007-0036-3. arXiv:math/0608399.

[L3] J. Sun and L. Yang, “Generalized Lagrangian Mean Curvature Flows in Almost Calabi-Yau Manifolds,” *J. Geom. Phys.* 117 (2017), 68-83. DOI: 10.1016/j.geomphys.2017.03.001. arXiv:1307.7854.

[L4] B. Lambert, J. D. Lotay and F. Schulze, “Ancient Solutions in Lagrangian Mean Curvature Flow,” *Ann. Sc. Norm. Super. Pisa Cl. Sci.* 22 (2021), 1169-1205. DOI: 10.2422/2036-2145.201901_016. arXiv:1901.05383.

[L5] J. D. Lotay, F. Schulze and G. Székelyhidi, “Ancient Solutions and Translators of Lagrangian Mean Curvature Flow,” *Publ. Math. IHÉS* 140 (2024), 1-35. DOI: 10.1007/s10240-023-00143-5. arXiv:2204.13836.

[L6] X. Li, Y. Luo and J. Sun, “Type II Singularities of Lagrangian Mean Curvature Flow with Zero Maslov Class,” *J. Funct. Anal.* 289 (2025), Article 111032. DOI: 10.1016/j.jfa.2025.111032. arXiv:2412.15880.

10.

Chen-Li 4.7 Wang 2001 Chen-Li 2002 Li-Yang 2014 Zhang 2015 Qiu-Sun 2019/2020

“ ” “ ”

4.7 “ ”

/ Chen-Li 2002

Kähler Qiu-Sun